Grade: Check Plus

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February 23, 2026
Statistics & Data Analysis
Problem set: assignment 1

a. What is the probability that a randomly selected customer withdraws at least ✓ Qa $60 from the ATM? There's a 45% chance

.15+ .25 + .05 = .45

b. Show by calculation that the expected value withdrawn from the ATM is $50 and that the standard deviation of the amount withdrawn from this ATM is $26.27. ✓ Qb

- $50= ∑x f(x)
  10 x .10 = 1
  20 x .15 = 3
  40 x .30 = 12
  60 x .15 = 9
  80 x .25 = 20
  100 x .05= 5
  1+3+12+9+20+5 = 50

- σ 2 =∑(x−μ) 2 f(x)

10-50= (-40)^2 = 1600	x .10 = 160
20-50= (-30)^2 = 900	x .15 = 135
40-50= (-10)^2 = 100	x .30 = 30
60-50= (10)^2 = 100	x .15 = 15
80-50= (30)^2 = 900	x .25 = 225
100-50= (50)^2 = 2500	x .05 = 125

  160+135+30+15+225+125= 690
- σ = square root of 690 = 26.267

c. The manager of this convenience score tells you that an expected value of $50 is not correct because it is impossible to withdraw $50 from the ATM. How would you answer the manager?

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✓ Qc

I would tell the manager that the mean of $50 is a probability prediction that represents an average amount to withdraw over time. It is not a specific value.

d. What is the probability that the amount withdrawn from the ATM will fall within

✓ Qd

one standard deviation of the mean?

There is a 45% chance

✓ Qe

e. What is the probability that the amount withdrawn from the ATM will fall within

two standard deviations of the mean?

There is a 100% chance

.10+.15+.30+.15+.25+.05= 1.00